

INTERNATIONAL STANDARD

Organic light emitting diode (OLED) displays – Part 2-1: Essential ratings and characteristics of OLED display modules



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2015 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

IEC Catalogue - webstore.iec.ch/catalogue

The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

IEC publications search - www.iec.ch/searchpub

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, replaced and withdrawn publications.

IEC Just Published - webstore.iec.ch/justpublished

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

Electropedia - www.electropedia.org

The world's leading online dictionary of electronic and electrical terms containing more than 30 000 terms and definitions in English and French, with equivalent terms in 15 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

IEC Glossary - std.iec.ch/glossary

More than 60 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

IEC Customer Service Centre - webstore.iec.ch/csc

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: csc@iec.ch.

INTERNATIONAL STANDARD

Organic light emitting diode (OLED) displays – Part 2-1: Essential ratings and characteristics of OLED display modules

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 31.260

ISBN 978-2-8322-2697-1

Warning! Make sure that you obtained this publication from an authorized distributor.

CONTENTS

FOREWORD	3
INTRODUCTION	5
1 Scope	6
2 Normative references	6
3 Organic light emitting diode (OLED) display modules	6
3.1 Types of OLED display modules	6
3.2 Principles and structure	6
3.3 Modes of operation	6
3.3.1 Addressing mode of operation	6
3.3.2 Optical mode of operation	6
3.4 Details of outline	7
3.4.1 Material and mechanical description	7
3.4.2 Connection types	7
3.4.3 Outline drawing and dimensions	7
3.5 Limiting values (absolute maximum rating system) over the operating temperature range, unless otherwise stated	7
3.6 Electrical and optical characteristics	7
3.7 Supplementary information	9
Table 1 – Limiting values of OLED modules	7
Table 2 – Electrical and optical characteristics of OLED modules	8

INTERNATIONAL ELECTROTECHNICAL COMMISSION

ORGANIC LIGHT EMITTING DIODE (OLED) DISPLAYS –**Part 2-1: Essential ratings and characteristics of OLED display modules****FOREWORD**

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 62341-2-1 has been prepared by technical committee 110: Electronic display devices.

The text of this standard is based on the following documents:

FDIS	Report on voting
110/651/FDIS	110/666/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 62341 series, published under the general title *Organic light emitting diode (OLED) displays*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

IEC 62341 consists of the following parts, under the general title *Organic light emitting diode (OLED) displays*:

- Part 1-1: Generic specifications
- Part 1-2: Terminology and letter symbols
- Part 2-1: Essential ratings and characteristics of OLED display modules
- Part 5: Environmental testing methods
- Part 5-2: Mechanical endurance testing methods
- Part 5-3: Measuring methods of image sticking and lifetime
- Part 6-1: Measuring methods of optical and electro-optical parameters
- Part 6-2: Measuring methods of visual quality and ambient performance
- Part 6-3: Measuring methods of image quality

ORGANIC LIGHT EMITTING DIODE (OLED) DISPLAYS –

Part 2-1: Essential ratings and characteristics of OLED display modules

1 Scope

This part of IEC 62341 describes the essential ratings and characteristics of organic light emitting diode (OLED) display modules.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Void.

3 Organic light emitting diode (OLED) display modules

3.1 Types of OLED display modules

The OLED display modules include the following types:

- passive matrix monochrome OLED display modules,
- passive matrix colour OLED display modules,
- active matrix colour OLED display modules.

3.2 Principles and structure

The OLED display modules consist of OLED display panels, electronic circuits, and usually bezels.

EXAMPLE A thin film transistor (TFT) (polycrystalline silicon, oxide, organic materials) active matrix display cell with electronic circuits and connector pins.

3.3 Modes of operation

3.3.1 Addressing mode of operation

The addressing mode of operation for OLED display modules includes the following types:

- passive matrix,
- active matrix, etc.

3.3.2 Optical mode of operation

The addressing mode of operation for OLED display modules includes the following types:

- illumination mode: for example top-emission, bottom-emission,
- number of colours,
- number of grey scales.

3.4 Details of outline

3.4.1 Material and mechanical description

The material and mechanical description of OLED display modules is as follows:

- substrate material: for examples glass, plastic, etc.,
- bezel material: for example plastic, metal, etc.

3.4.2 Connection types

The connection types of OLED display modules are as follows:

- connector,
- flex cable,
- connection pins, etc.

3.4.3 Outline drawing and dimensions

The outline drawing and dimensions of OLED display modules are as follows:

- overall dimensions,
- bezel dimensions,
- active display area and display centre.

3.5 Limiting values (absolute maximum rating system) over the operating temperature range, unless otherwise stated

The following limiting values should be specified in Table 1.

Table 1 – Limiting values of OLED modules

Subclause	Characteristics	Symbol	Requirements		Unit
3.5.1	Operating temperature	T_{op}	Min.	Max.	°C
3.5.2	Operating humidity	H_{op}		Max.	%
3.5.3	Storage temperature	T_{stg}	Min.	Max.	°C
3.5.4	Storage humidity	H_{stg}		Max.	%
3.5.5	Supplied voltages				
	Supply voltage for logic drive	$V_{DD} - V_{SS}$	Min.	Max.	V
	Supply voltage(s) for EL	V_{EL}	Min.	Max.	
3.5.6	Input signal voltages	V_{IN}	Min.	Max.	V

3.6 Electrical and optical characteristics

The following characteristics should be specified in Table 2.

Table 2 – Electrical and optical characteristics of OLED modules

Subclause	Characteristics	Condition at $T_{op} = 25\text{ °C}$ unless otherwise specified	Symbol	Requirements		Unit
3.6.1	Supplied voltages Supply voltage for logic drive Supply voltage(s) for EL		$V_{DD} - V_{SS}$ V_{EI}	Min. Min.	Max. Max.	V
3.6.2	Input signal voltages High level input signal voltage Low level input signal voltage Input analogue video signals (where appropriate)		V_{IN} V_{INH} V_{INI} V_{VID}	Min. Min. Min. Min.	Max. Max. Max. Max.	V
3.6.3	Operating frequency (where appropriate) Frame frequency		f_{op} f_{FRM}	Min.	Max.	Hz
3.6.4	Supply currents	Conditions chosen to achieve maximum supply current, e.g. operating supply voltage, display pattern, etc. as appropriate	I_{tot} or I_{DD} and/or I_{EE}		Max.	mA
3.6.5	High level input signal current (where appropriate)		I_{INH}		Max.	mA
3.6.6	Low level input signal current (where appropriate)		I_{INL}		Max.	mA
3.6.7	Dark-room contrast ratio	Specified measuring method and conditions	$DRCR$	Min.		–
3.6.8	Ambient contrast ratio	Specified measuring method and conditions	ACR	Min.		–
3.6.9	Luminance (where appropriate) Luminance uniformity (where appropriate) Color uniformity (where appropriate)	Specified measuring method and conditions	L L_{uni} $\Delta u'v'$	Min. Min. .	Max.	cd/m ² – –
3.6.10	Viewing angle range	Specified definition of viewing direction and specified contrast ratio	θ_V and θ_H	Min.	Max.	°
3.6.11	Half luminance lifetime	Specified measuring method and conditions	T_{50}	Min.		h
3.6.12	Power consumption	Specified measuring method and conditions	P		Max.	W
3.6.13	Reflectance (where appropriate)	Specified measuring method and conditions	ρ		Max.	–
3.6.14	Chromaticity of white (x,y) (where appropriate) Chromaticity of red (x,y) (where appropriate) Chromaticity of blue (x,y) (where appropriate) Chromaticity of green (x,y) (where appropriate)		x_W, y_W x_R, y_R x_B, y_B x_G, y_G	a a a a	a a a a	–
3.6.15	Image sticking (where appropriate)	Specified measuring method and conditions	IS		Max.	–
^a Items of value (i.e., minimum, maximum, typical or average) are determined in a detail specification.						

3.7 Supplementary information

- 3.7.1** Timing characteristics, timing of logic voltages and data/format interface specification.
 - 3.7.2** Supply voltages sequence condition, where appropriate.
 - 3.7.3** Operating voltage range, if appropriate, as a function of temperature at specified contrast ratio.
 - 3.7.4** Handling and operating information.
 - 3.7.5** Precautions with respect to electrostatic discharges.
 - 3.7.6** Precautions of installation: mechanical and/or electrical.
 - 3.7.7** Safety information.
 - 3.7.8** Characterization of diffuse and specular reflectance and transmittance.
-

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch